



Technical Service Bulletin

Technical Service Bulletin: TSB180021

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Two Stage Fuel Air Bleed Valve

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK19 CM850 MCRS
- QSK19 CM2150 MCRS
- QSK38 CM850 MCRS
- QSK38 CM2150 K106
- QSK50 CM850 MCRS
- QSK50 CM2150 K107
- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS

Note : If an application uses a MCRS CM850 or MCRS CM2150 for Tier 2 or Tier 3 and needs to be upgraded to Tier 4 Stage 2 Filtration Package, the two stage air bleed valve will be needed.

Issue

Symptom:

- Hard Starting: Single stage air bleed valve equipped engines may be difficult to prime after fuel filter change.
- External leakage from stage 1 industrial pro fuel filter top (fill cap): Single stage air bleed valve equipped Tier 2 and Tier 3 MCRS engines upgraded to Tier 4 stage 2 filtration packages may experience stage 1 industrial pro fuel filter o-ring malfunction caused by increased volume of stage 2 filter.

Root Cause:

- Trapped air in fuel system after fuel filter change can prevent the fuel lift pump from delivering adequate fuel during cranking.
- Possible external fuel leakage from stage 1 industrial pro fuel filter top (fill cap): stage 1 fuel filter cap o-ring can be dislodged by a pressure increase after engine shutdown.

Verification

The new two stage air bleed valve is recommended to be used in the following scenarios:

- If engine family meets criteria in tables below, and engine is difficult to start.
- During engine rebuild, affected engines should be upgraded with 2 stage air bleed valve.
- 2 stage air bleed valve is required when upgrading an engine with a stage 2 filtration package.
- When single stage air bleed valve requires replacement, upgrade to 2 stage air bleed valve is recommended.

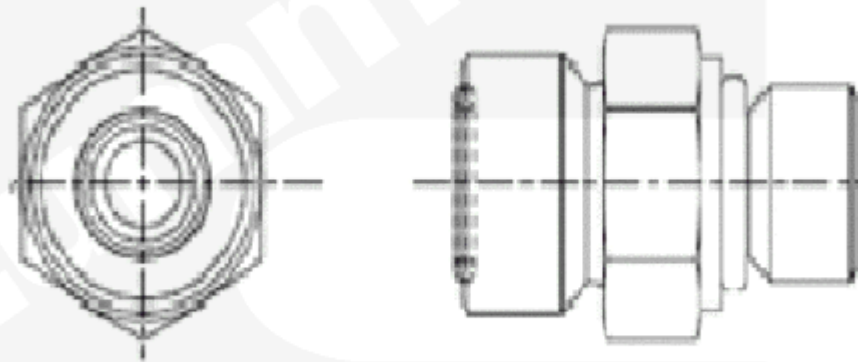
Resolution

Use tables in Service Parts Availability section below to order 2 stage air bleed valve.

Description of Change

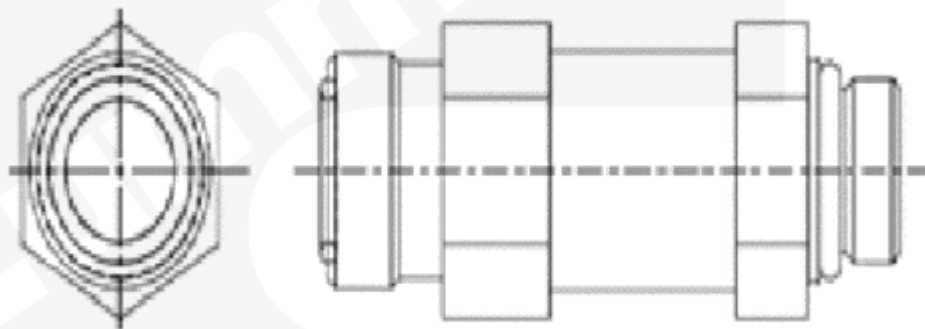
MCRS fuel system changed from a single stage check valve to a 2 stage check valve.

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Figure 1, Single Stage Air Bleed Valve, Part Number 4928063



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Figure 2, 2 Stage Air Bleed Valve, Part Number 4379736

Service Parts Availability

Note : See table 1 for fuel systems that require additional plumbing for new two stage air bleed valve.

Note : In Table 1, the marine application does not use the two stage air bleed valve; construction and generator set use the additional fuel drain hose.

Service parts available. See Table 1 below for part numbers.

Table 1, QSK19 Service Parts for Application: Construction, Generator Set, Marine

Engine Family	Application	Option	Part Description	Existing Part Number	New Part Number	Fuel Drain Hose Plumbing	Refer to Procedure 006-020 (/qs3/pubsys2/xml/en/procedures/20/20-006-020.html)
QSK19 CM850	Construction	FD 4701	Air Bleed Valve	4928063	4379736	4334075	
	Generator set						
QSKTA19	Marine	FD 4703			-	-	

Service parts available. See Table 2 below for part numbers.

Table 2, QSK38, QSK50 Service Parts for Application: Construction, Generator Set

Engine Family	Application	Option	Part Description	Existing Part Number	New Part Number	Fuel Drain Hose Plumbing	Refer to Procedure 006-020 (/qs3/pubsys2/xml/en/procedures/28/28-006-020.html)
QSKTA38	Generator set	FD 6735	Air Bleed Valve	4928063	4379736	-	
	Construction	FD 6738 FD 6739					
QSKTA50	Generator set	FD 6707 FD 6708					
QSKTA50	Construction	FD 6740 FD 6741					

Service parts available. See Table 3 below for part numbers.

Table 3, QSK60 Service Parts for Application: Construction, Generator Set

Engine Family	Application	Option	Part Description	Existing Part Number	New Part Number	Fuel Drain Hose Plumbing	Refer to Procedure 006-020 (/qs3/pubsys2/xml/en/procedures/56/56-006-020.html)
QSKTA60	Generator set	FD 6709	Air Bleed Valve	4928063	4379736	-	
QSKTTA60	Generator set	FD 6744 FD 6745					
QSKTA60	Construction	FD 6745 FD 6744					

Part Compatibility

When applying Tier 4 Stage 2 filtration package on Tier 2 or Tier 3 engines, the 2 stage air bleed valve **must** be installed.

Production Status

Implemented for production. See Table 4

Table 4, Production Information

ESN First	Build Date ¹	Plant
33205247	06 July 2015	DAV-Daventry Engine Plant
37270347	16 January 2017	CIC/SEP – Seymour Engine Plant

¹ Engine build date can be found on the engine data plate.
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Document History

Date	Details
2018-2-9	Module Created

Last Modified: 08-Mar-2018
